Process Validation

Science

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Process Validation (A09764)

Short Title:	Process Validation		
Faculty	Science and Computing		
Department	Science		
Cluster	Quality, Regulatory and Compliance		
Author	BWHELAN		
Credits	5	**Level:**	Intermediate

Description of Module / Aims

This module describes the importance and practices of validation and technology transfer. Both areas are linked in terms of design and commissioning of a process, piece of equipment or activity within the pharmaceutical industry. Other topics covered include Computer Systems Validation, Process Analytical Technology (PAT) and Quality by Design (QbD).

Programmes

`PROC-0005`	BSc in Good Manufacturing Practice and Technology (`WD_SGMPT_D`)	1 / 1 / M

Indicative Content

- Process validation: general concepts and documentation required; DQ, IQ, OQ, PQ; validation strategy
- Principles of Quality by Design (QbD), life cycle approach to process validation, worst case testing
- Clean Room qualification, manufacture of sterile medicinal products
- Process Analytical Development (PAT) development, change management systems, risk management methods and tools
- Validation of process water, purified water systems, water for Injection
- Commissioning, Factory Qualification, Factory Acceptance Testing (FAT), Site Acceptance Testing (SAT)
- Description of solid dosage forms, transdermal delivery systems, freeze-dried products, inhalation products; medical device development
- Computer systems validation, GAMP guide, Quality Systems and Compliance, 21 CFR Part 11

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Interpret validation guidance documents used within the pharmaceutical/medical device sectors.
2. Develop validation protocols identifying critical parameters and acceptance criteria.
3. Complete FATs and SATs and enhance process understanding.
4. Appraise the application of risk management to a range of facilities, processes and products.
5. Plan for quality system application throughout product life cycle.

Learning and Teaching Methods

- Lectures.
- Case studies.
- Class participation.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
In-Class Assessment	50%	1,2,4
Group Project	50%	3,5

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter. Lacking in technical ability/ skill and understanding to carry out practical/professional tasks.

40%-49%: Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on practical/professional tasks.

50%-59%: Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out practical/professional tasks.

60%-69%: Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design when carrying out practical/ professional tasks.

70%-100%: Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject. Demonstrate imaginative approach, excellent technical ability, awareness, and aptitude in carrying out and reporting on practical/professional tasks.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture		24
Independent Learning		111

Essential Material

- "Food and Drugs Administration." http://www.fda.gov/
- Nash, R.A. and A.H. Wachter. _Pharmaceutical Process Validation_. 3rd. New York: Marcel Dekker, Inc, 2003.

Supplementary Material

- "International Society for Pharmaceutical Engineering." http://www.ispe.org/
- "Journal of Validation Technology." http://www.ivtnetwork.com/jvt-journal